

BHARATH KRISHNAN S

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Professional Summary

CSE undergraduate specializing in Machine Learning, Deep Learning, Computer Vision, and NLP, with hands-on experience building, evaluating, and deploying AI models across object detection, behavioural analysis, predictive maintenance, and explainable AI.

Education

Bachelor of Engineering in Computer Science and Engineering (B.E. CSE) 2023 – 2027
Chennai Institute of Technology and Applied Research, Chennai CGPA: 9.05 / 10

Work Experience

Machine Learning Intern — Saiket Systems (Remote) Mar – Apr 2026

- Engineered Machine Learning models (Scikit-learn, TensorFlow) for classification/regression with hyperparameter tuning and accuracy, F1-score model evaluation.
- Performed EDA, data preprocessing, and feature engineering to improve model performance.

Front-End Developer Intern — SSV's Traders (Hybrid) May 2024

- Built responsive UI (HTML, CSS, JS), improving compatibility by 30% in a 4-member Agile team.
- Integrated REST APIs for dynamic data exchange, reducing data-fetch latency by 20%.

Data Science Intern — Cognifyz Technologies (Remote) Mar – Apr 2026

- Applied Machine Learning and statistical analysis to real-world datasets using Python and Scikit-learn.
- Strengthened model development, data preprocessing, and analytical problem solving.

Projects

RailMind AI – Railway Safety & Behaviour Detection 2026

Python · YOLOv8 · BiLSTM · LangGraph · PyTorch

- Built a CV pipeline (YOLOv8 + ByteTrack + Pose) feeding a 30s BiLSTM classifier, targeting 92% accuracy at sub-500ms latency.
- Orchestrated a three-tier LangGraph multi-agent system for 0–100 risk scoring and automated escalation.

Traffic Violation Detection – Flipkart Gridlock 2.0 2026

Python · YOLOv11 · ByteTrack · EasyOCR · Flask

- Engineered a 10-violation system with a temporal rule engine, reaching 0.91 mAP on plate and 0.89 on helmet detection.

- Generated tamper-evident SHA-256 evidence with live MJPEG monitoring and a React analytics dashboard.

FactoryGuard AI – IoT Predictive Maintenance 2026

Python · XGBoost · LightGBM · SHAP · Flask

- Built an XGBoost/LightGBM/RF ensemble with consensus voting, achieving 0.75 PR-AUC on imbalanced failure data.
- Engineered rolling/lag features and SHAP explainability with live IoT ingestion and audit-trail persistence.

VisionSpec QC – Explainable PCB Defect Inspection 2026

TensorFlow · ResNet50 · Grad-CAM · FastAPI

- Developed a transfer-learning classifier (ResNet50/MobileNetV2) with Grad-CAM heatmaps for explainable detection.
- Optimized models via quantization and served low-latency FastAPI inference with live webcam streaming.

Technical Skills

- Programming Languages:** Python, C, Java, SQL
- Machine Learning:** Scikit-learn, XGBoost, LightGBM, Random Forest, Ensemble Learning, SHAP (Explainable AI), Hyperparameter Tuning
- Deep Learning & CV:** PyTorch, TensorFlow, Keras, CNN, RNN, LSTM / BiLSTM, Transfer Learning (ResNet50, MobileNetV2), YOLOv8 / YOLOv11, ByteTrack, Grad-CAM, OpenCV, EasyOCR
- NLP & GenAI:** Transformers, Hugging Face, NLP, Sentiment Analysis, LangGraph (Multi-Agent), LLM Integration
- Data Science & Tools:** Pandas, NumPy, Matplotlib, EDA, Feature Engineering, Model Evaluation (F1, mAP, PR-AUC), Jupyter, Streamlit, Docker, Git/GitHub
- Deployment & Databases:** FastAPI, Flask, REST APIs, SQLite, PostgreSQL, MongoDB
- Areas of Interest:** Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Science

Competitive Programming

LeetCode: Rating 1600, Problems Solved 450+ sbharathkrishnan

CodeChef: Rating 1224, Problems Solved 200+ sbkrishnan27

Codeforces: Rating 1030, Problems Solved 50+ sbharathkrishnansbk

Achievements & Activities

- TCS CodeVita Season 12 & 13** – Achieved Global Rank **6599 & 5107** worldwide.
- National **Hackathon Finalist** – Shortlisted among top teams at HackOrbit and the **NIT Trichy** FinTech Hackathon.
- NPTEL Elite Certification – Ethical Hacking by IIT Kharagpur with a 73% score.
- Secured **1st** place in the **ML Ware Hackathon Championship** conducted by **IIT Varanasi** on Kaggle.

Certifications & Coursework

- AWS :** Cloud Practitioner – Generative AI Foundations – ML Foundations
- IBM :** AI Essentials – Web Development Essentials
- Cisco :** Intro to Networks – Cybersecurity Essentials – Data Analytics Essentials
- NPTEL (IIT Kharagpur):** Ethical Hacking – Elite Certification
- Coursera :** DSA – Python for Data Science – Google Cybersecurity